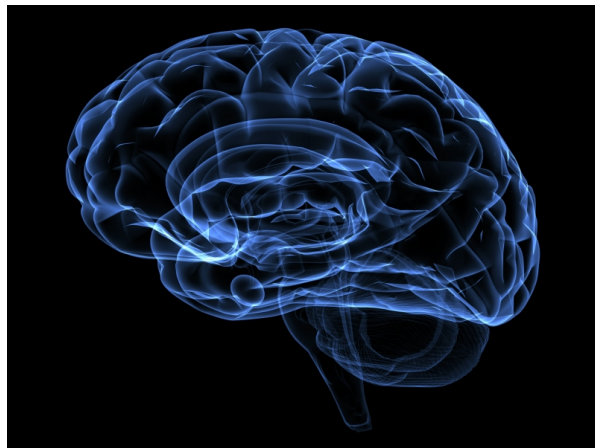




Pediatric Neuromusculoskeletal Disorders Part 2



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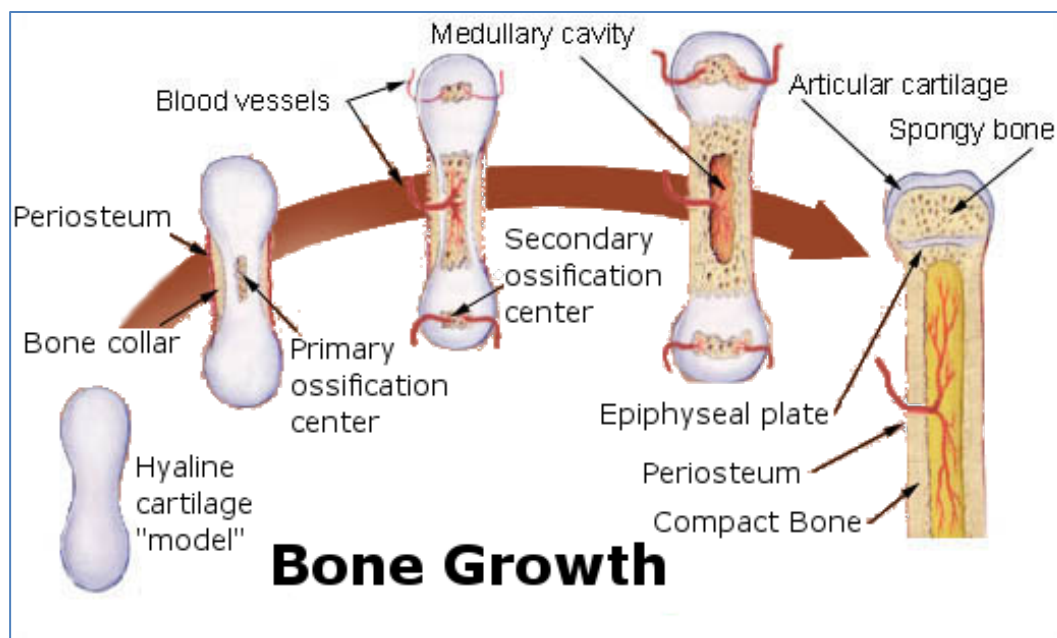
Objectives

At the end of this module the learner should be able to:

- Recognize and describe the following pediatric orthopedic problems:
 - Talipes Equinovarus
 - Metatarsus Adductus
 - Calcaneal Apophysitis
 - Femoral Anteversion
 - Internal Tibial Torsion
 - Tibia Vara
 - Osgood-Schlatter Disease
 - Developmental Dysplasia of the Hip
 - Transient Synovitis
 - Legg-Calve-Perthes Disease
 - Slipped Capital Femoral Epiphysis
 - Scoliosis
 - Kyphosis
 - Torticollis
 - Spondylolysis
 - Spondylolisthesis
 - Diskitis
 - Radial Head Subluxation
- Identify the most common presentations; and describe the diagnostic tests and treatment for the above.

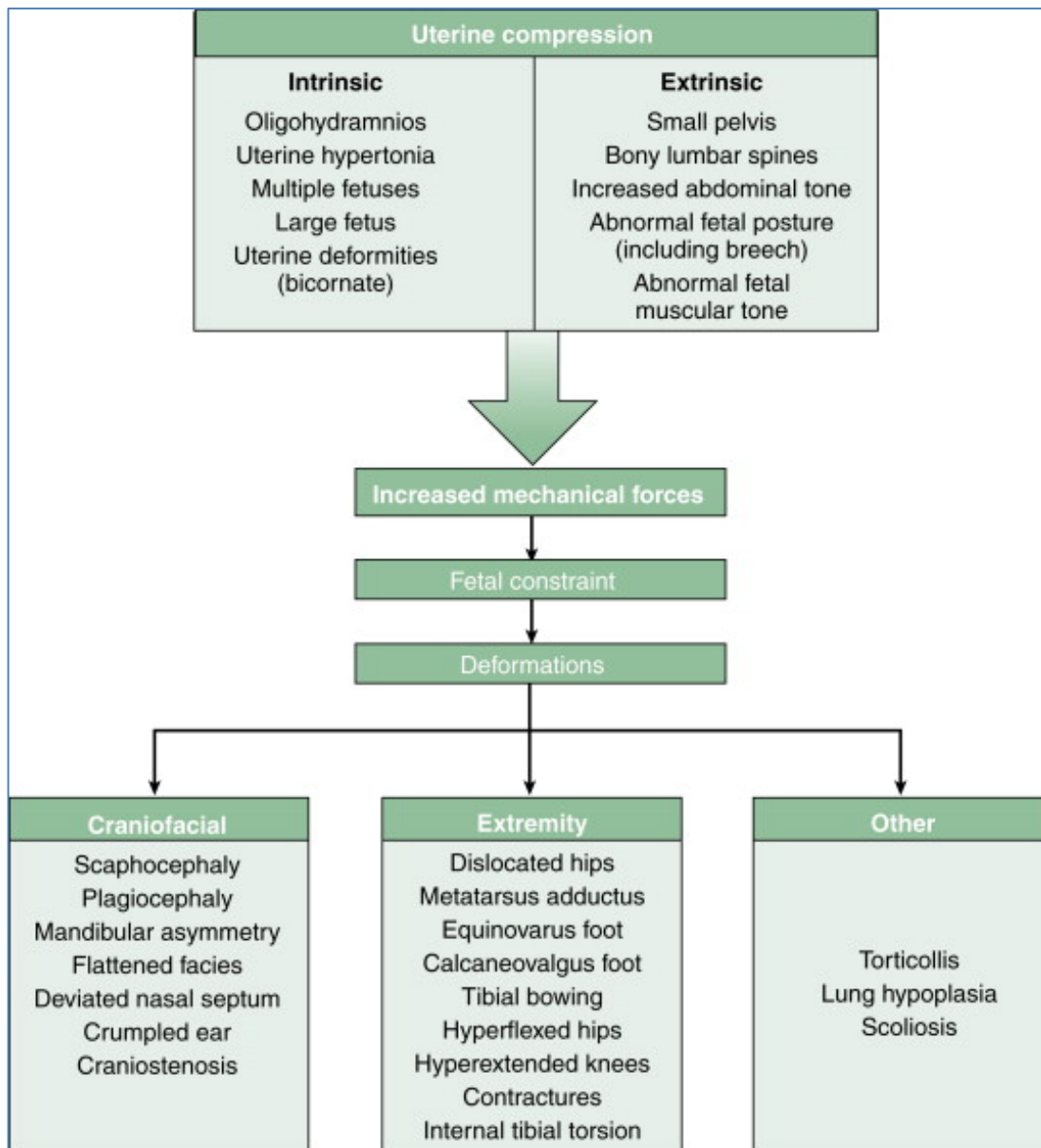
Pediatric Orthopedic Considerations

Pediatric patients are not simply small adults. They have different anatomy and physiology than adults do. In pediatric patients, it is important to consider the growth and development of bones. The ends of long bones in young patients contain a higher proportion of cartilage, allowing for a unique vulnerability to trauma and infection.



It is also important to consider the in utero position of the fetus. The position of the fetus as it is growing and developing can affect the angular and torsional alignment of the musculoskeletal system. Even after birth, newborns continue to hold their hips externally rotated, ankles inverted, forefoot adducted, and naturally revert to the flexed posture they were in utero.





The Foot

Posturing is a habitual position in which the infant holds his or her foot. The passive range of motion in posturing is normal.

A **deformity** appears similar to posturing; however, the passive range of motion is restricted.



Talipes Equinovarus (Club Foot)

Talipes equinovarus, or “club foot”, involves the **entire leg** and is found in 1 in 1000 newborns. Most cases are congenital (75%), however there are also teratologic (myelomeningocele, arthrogryposis), and in-utero positional causes.

50% of the cases have clubbing bilaterally. Talipes equinovarus is caused by abnormal tarsal interaction resulting in generalized limb hypoplasia and thus the shortening of the foot. Patients present with **hindfoot equinus** (heel pulled upward) and **varus, forefoot adduction**, and varying degrees of rigidity.

Diagnosis is made clinically (based on physical examination). X-rays are not needed (remember, the navicular bone ossifies around 3 to 4 years of age, so x-rays prior to that time will not be useful).

Treatment involves serial casting of the foot and distal lower extremity to slowly bring the limb into a normal alignment. Surgical Achilles tenotomy is often necessary as well.



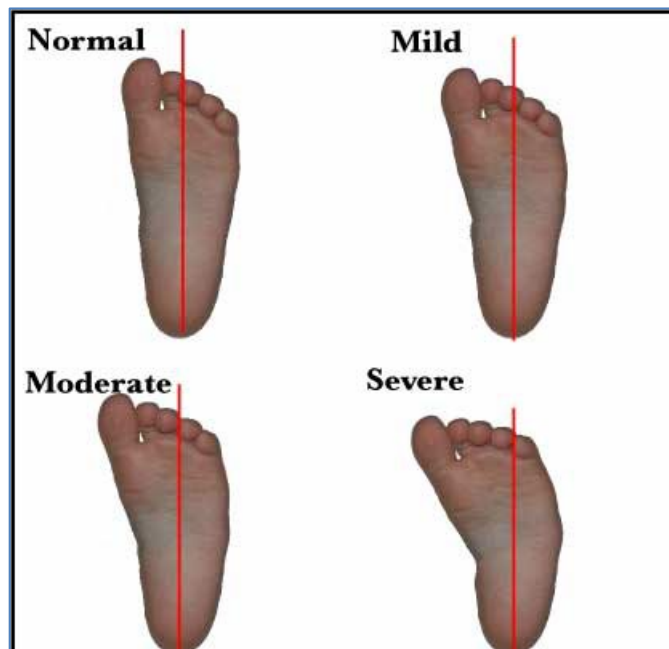
Metatarsas Adductus

Metatarsas Adductus is the **most common** foot disorder in infants. Patients with this disorder have a **convexity of their lateral foot**. This is caused by **in utero** positioning and is more common in primigravid. 50% cases are presented with bilateral convexity of the lateral foot and about 2% have DDH (developmental dysplasia of the hip)



Diagnosis is made clinically (based on physical examination).

Treatment: Most cases (about 90%) **resolve spontaneously**. If the patient is older than 2 years of age, and/or if they have significant pain or difficulty ambulating, then serial casting, bracing, or surgery are considered.



Polydactylyl and Syndactyly

Patients with **polydactyly** have **extra digits** and patients with **syndactyly** have the **fusion of digits**. Syndactyly is more common than polydactyly.

Both are usually **benign cosmetic** problems and may be associated with other syndromes (listed below):

POLYDACTYLY	SYNDACTYLY
Ellis-van Creveld syndrome	Apert syndrome
Rubinstein-Taybi syndrome	Carpenter syndrome
Carpenter syndrome	de Lange syndrome
Meckel-Gruber syndrome	Holt-Oram syndrome
Polysyndactyly	Orofaciodigital syndrome
Trisomy 13	Polysyndactyly
Orofaciodigital syndrome	Fetal hydantoin syndrome
	Laurence-Moon-Biedl syndrome
	Fanconi pancytopenia
	Trisomy 21
	Trisomy 13
	Trisomy 18



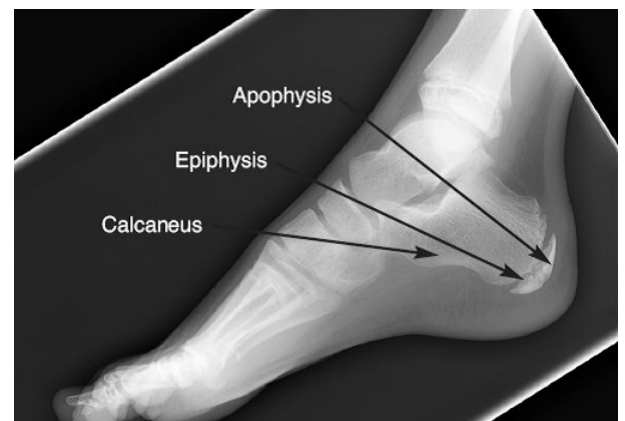
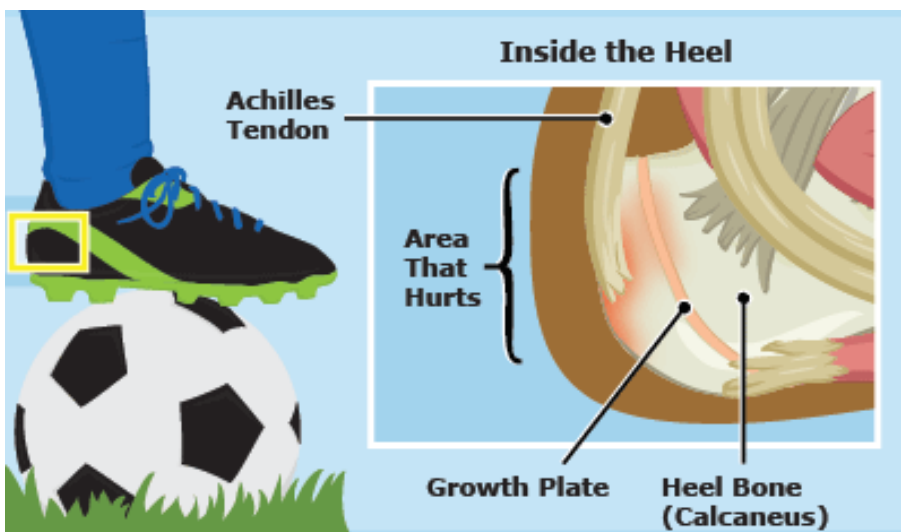
Calcaneal Apophysitis (Sever Disease)

Calcaneal Apophysitis is a common cause of **heel pain in active young people**. The average age for females is 9 years and for males it is 11 to 12 years. Calcaneal apophysitis occurs bilaterally about 60% of the time.

This disorder affects the calf muscle, Achilles tendon, and can result in **microfracture of the calcaneal apophysis**.

The classic history of present illness (HPI) for calcaneal apophysitis is a **young athlete who presents with heel pain with activity and who reports that the pain decreases with rest**. Patients may present with limping. There is rarely any swelling. On physical examination, there is **tenderness to palpation of the posterior calcaneus** and a hypertonic Achilles.

Treatment includes rest, ice, NSAIDs, stretching, and heel wedges can be considered.



The Distal Lower Extremity

Children are most often brought in to the pediatrician for issues related to the lower extremity and the knee out of parental concern.

Parents may be concerned that their children are not developing normally. In-toeing, out-toeing, physiologic “bowlegs”, and “knock knees” are common reasons parents bring their child in. Most of these are **physiologic**, bilateral, and should **not** cause limp or pain. Generally this should resolve with normal growth.

Of Note: At birth the femur and tibia are internally rotated about 30 degrees. The bones slowly externally rotate to a



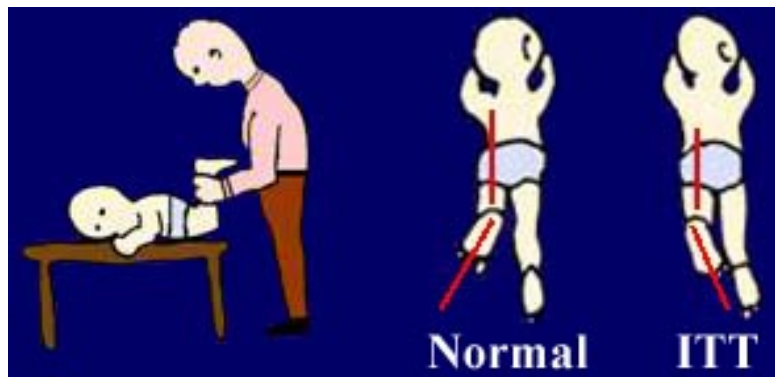
In-Toeing

The most common causes of in-toeing are **metatarsus adductus** in babies less than 1-year old, **internal tibial torsion** in toddlers ages 1-2 years old, and **femoral anteversion** in school age kids above 2 years of age. (*Notice how the problem moves from distal to proximal up the leg as the child ages*).



Internal Tibial Torsion

Toddlers generally present with **in-toeing**. On the physical exam, we **measure the thigh-foot angle** by having the patient lay prone with knee flexed to 90 degrees. Observe the angle that the foot makes with the femur (*see image below*).



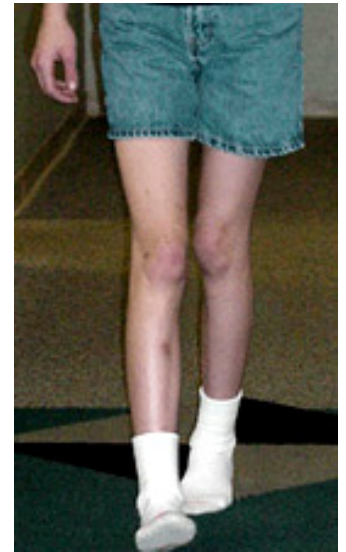
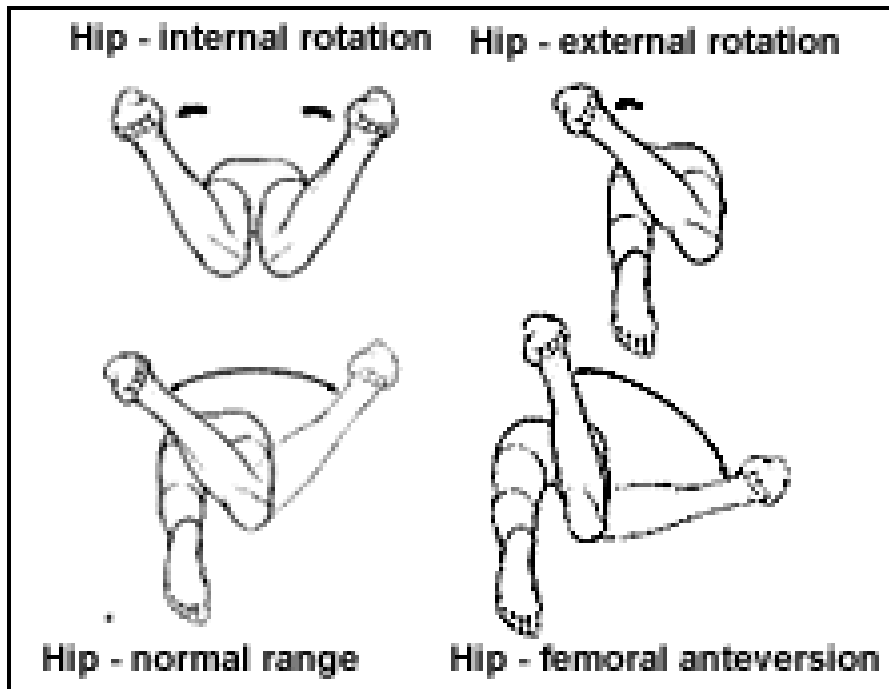
The treatment for this is **reassurance** and follow up. Generally, internal tibial torsion will self-resolve by 7-8 years of age. It rarely requires surgery (<1% of cases). Children who have significant pain or difficulty ambulating (which is very rare with ITT) should be referred to orthopedics.

Femoral Anteversion

Patients with femoral anteversion will generally **be females ages 4-6 years** who present with **in-toeing**. These patients often **sit in the “W-position”** (*see image.*) Females are two times more frequently affected compared to males.



On the **physical exam**, the entire leg appears internally rotated, giving the appearance of “**kissing knee-caps**” (*see image below*). For the exam, the patient should lay prone with knees flexed to 90 degrees, then allow distal leg to move laterally.



The **treatment** for this is **reassurance** and follow up. The patient should also **avoid W-sitting** (encourage sitting cross-legged). This typically resolves by 7-8 years of age, and, like ITT, rarely requires surgery (<1% of cases). Children who have significant pain or difficulty ambulating (which is very rare with femoral anteversion) should be referred to orthopedics.

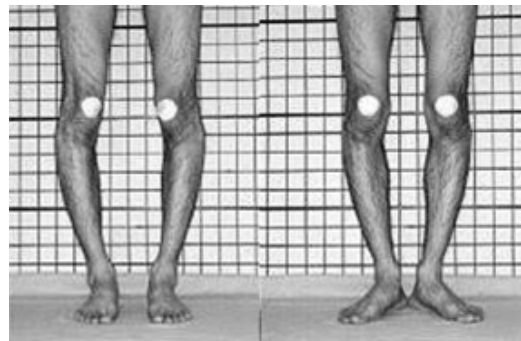
Expert Tip

Dr. Hendriksz encourages worried parents to watch professional basketball games and note how many of the players have in-toeing. In-toeing does not typically cause major concerns.



Out-Toeing

External Tibial Torsion is the **most common** cause of out-toeing. Remember that physiologically, the tibia rotate externally as children age, so external tibial torsion can worsen with age. This is often related to **in utero** positioning. The treatment for this condition would be **observation and reassurance**. If there is a significant dysfunction (pain or gait abnormalities) or cosmetic concerns, surgery is considered.

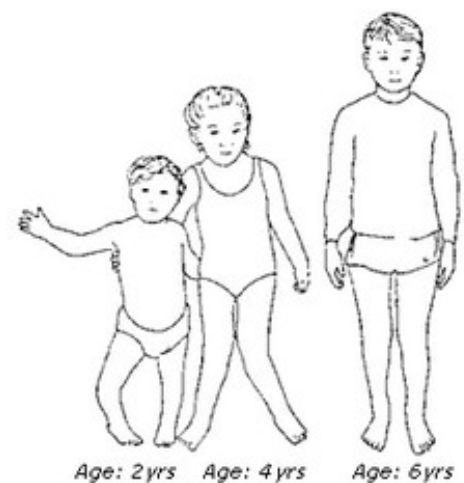


Genu Varum and Genu Valgum

Most individuals have the **normal physiologic variation**:

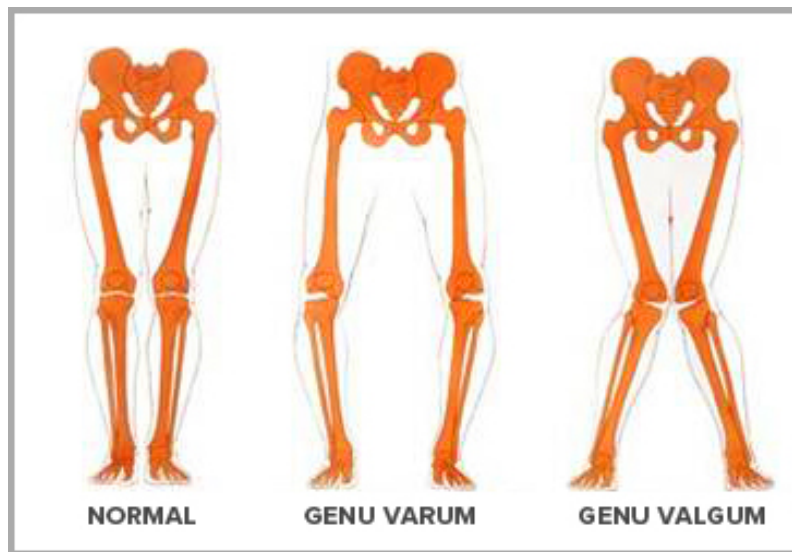
- Infants through toddler age have **genu varum**
- School-aged children (ages **4 years** – 8 years) have **genu valgum**
- Adolescents have a more typical normal adult alignment

For individuals presenting with genu varum or genu valgum, it is important to assess their overall height. If the child is less 2 standard deviations below the mean, he or she might have skeletal dysplasia.

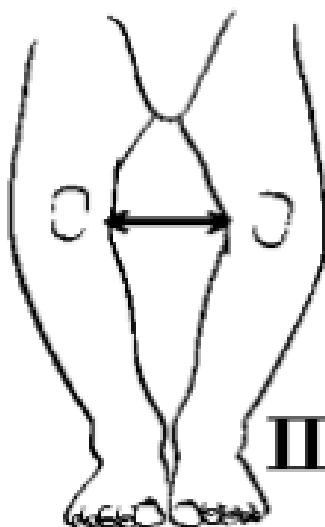


It is also important to closely investigate and monitor their diet and activities. Vitamin D deficiency can lead to **Rickets** or bone pain, dental deformities, impaired growth, and skeletal deformities.

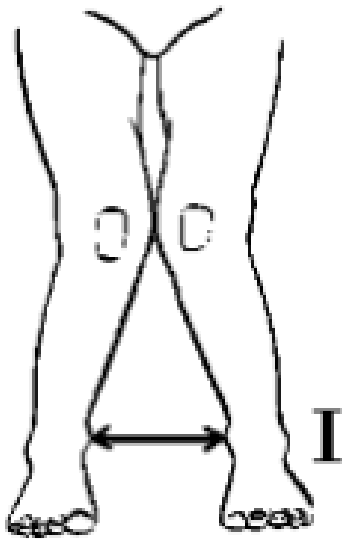
Note: The age at which a child becomes vitamin D deficient predicts how their Rickets will present (genu varum if they are under 2 years of age, genu valgum if they are older)



Children with **genu varum** are most commonly older than **18 months** and resolves by **2-3 years of age**. To diagnose this condition, we **measure the distance between the medial femoral condyles** (see *image below*) with the medial malleoli touching to rule out **tibia vara** (Blount disease). The treatment for genu varum is **reassurance**. Surgery is considered for children with severe deformities, gait dysfunction, and pain.



Children with **genu valgum** are most commonly **3 to 4 years old** and resolves by **5 to 8 years** of age. To diagnose this condition, we **measure distance between the two tibial medial malleoli** with the knees touching (*see images below*). The treatment for this genu valgum is **reassurance**. Surgery is considered for severe deformities, gait dysfunction, and pain.



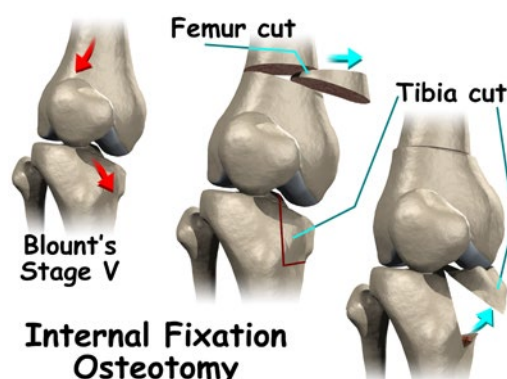
Tibia Vara (Blount Disease)

Tibia Vara is due to an **abnormal growth of medial aspect of proximal tibial epiphysis**. The underlying etiology of this disease is unknown. It is most commonly seen in the toddler age (1 to 3 years) in African Americans, females, and obese children. 80% of patients with tibia vara have a bilateral disorder, and **do not experience pain due to their condition**. A less common variation of tibia vara presents in adolescence. Those patients are most commonly African American, male, and markedly obese. About 50% of patients with adolescent tibia vara have a bilateral deformity, and many **experience pain**. There is also a third form while occurs in school-aged children (4 to 10 years of age).



To **diagnose** tibia vara, **weight-bearing AP and lateral x-rays** of both **lower extremities** are obtained. The x-ray will demonstrate fragmentation, **wedging**, and **beak deformities** of the **proximal medial tibia** (*see images*).

The **treatment** for those younger than 3 years old is to give **immediate orthotics**. **Surgery** is performed for those older than 4 years of age. Patients receive a proximal tibial valgus osteotomy with fibular diaphyseal osteotomy.

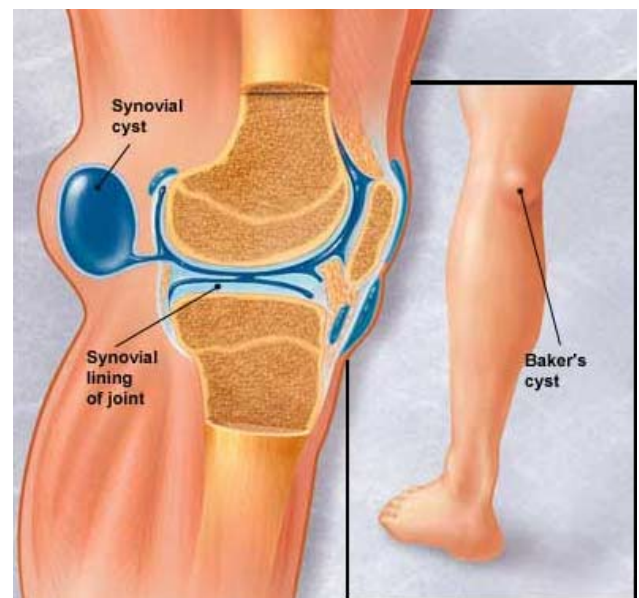


The Knee

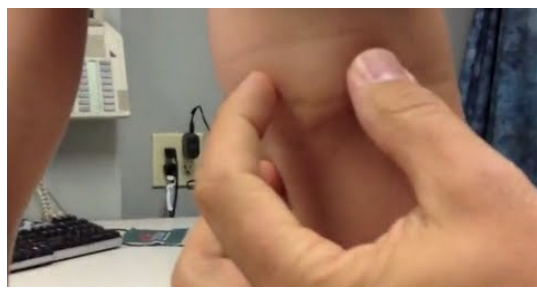
Popliteal Cyst (Baker's Cyst)

A popliteal cyst is common in the **middle childhood** years. It is caused by **distension of the gastrocnemius & semimembranous bursa along the posteromedial aspect** of the knee by synovial fluid. In adults, they are most commonly due to meniscal tears.

Pediatric patients tend **not to have pain** associated with popliteal cysts, and the **cysts are benign**. The cysts typically **resolve spontaneously**, though it may take a few years.



The treatment for these patients is **reassurance**. Surgery is considered if the cyst is progressive or limits mobility.



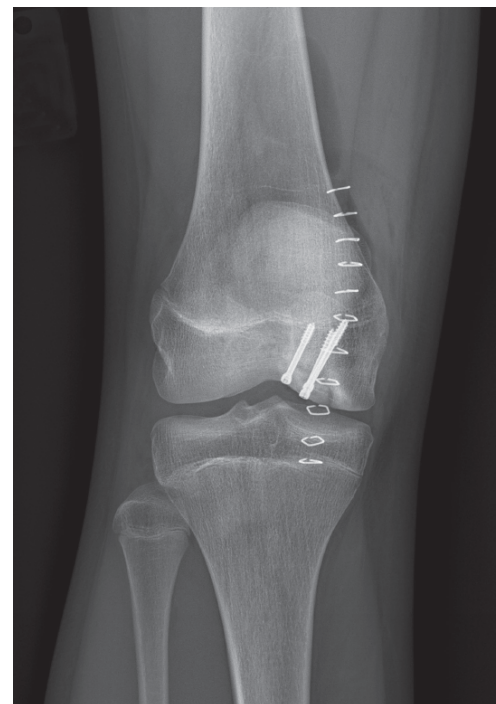
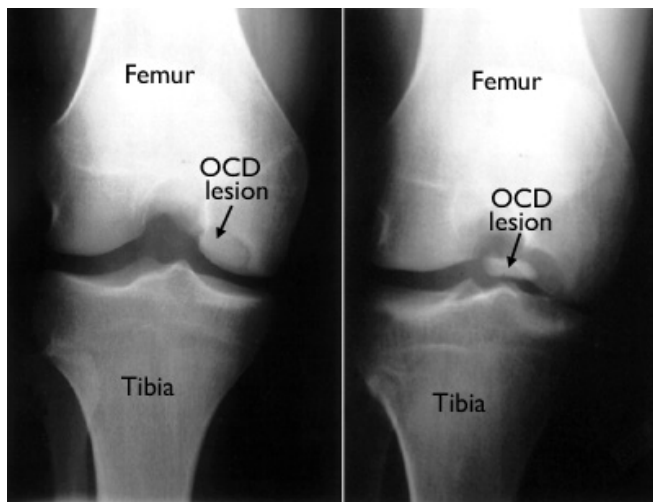
Osteochondritis Dissecans

Osteochondritis Dissecans most commonly affects the **knee**. It occurs when an **area of bone adjacent to the articular cartilage suffers a**

vascular insult and separates from the adjacent bone. This occurs most commonly at the lateral aspect of the medial femoral condyle.

Patients present with **knee pain and swelling.** The diagnosis is done with an **MRI.**

Treatment of the patient depends on the degree of separation of the articular cartilage. In young patients with **intact articular cartilage**, revascularization and healing with **rest** is considered. In older children with **damage to the cartilage**, **surgery** is considered.

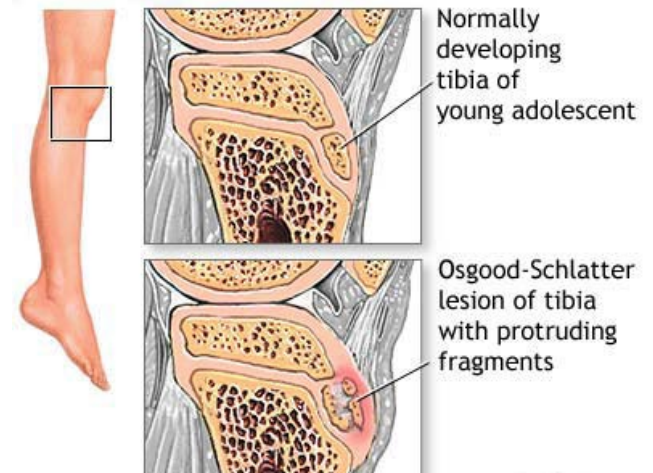


Osgood-Schlatter Disease

Osgood Schlatter Disease is most common in **boys 13-14 years of age**, and is also seen in females around 11 years of age. The disease usually occurs after a growth spurt. The disease results from the contracting quadriceps muscle causing **microfracture or partial avulsion** fractures in the ossification centers of the tibial tubercles.

Patients present with **knee pain at the insertion of the patellar tendon on the tibial tubercle**. The physical exam may show **tenderness and edema over the tibial tubercle**.

The treatment for this disease would be for the patient to rest and modify their activities to prevent excessive stress on the tibial tubercle. NSAIDs, ice, stretching, and lower extremity strengthening are also recommended. Symptoms will generally last 1-2 years.



Complications for this disease include **bony enlargement of the tibial tubercle and avulsion fractures**.

Patellofemoral Pain syndrome

Patellofermoal pain syndrome is common in **adolescent female athletes**. Patients present with **anterior knee pain, worse with activity** and stairs, and soreness after sitting in one position for extended time. The pain can be described as a **grinding sensation** under the kneecap.

On the physical exam, there is **tenderness to palpation and compression of**



patellofemoral joint with the knee extended. The hip musculature tends to be weak and there is often poor flexibility in the lower extremities. There is also generally no edema.

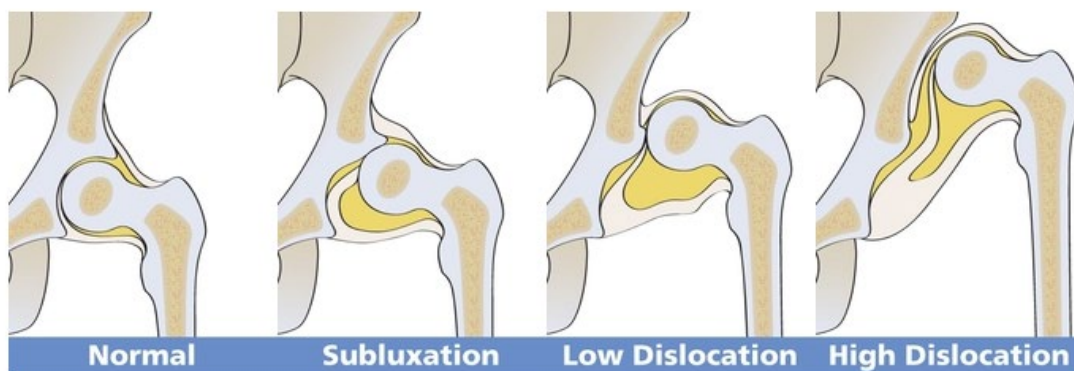
The treatment of this syndrome involves **strengthening the hip girdle** and increasing lower extremity flexibility. NSAIDs, rest, and ice are also recommended.

The Hip

Developmental Dysplasia of the Hip (DDH)

Hips at birth may be dislocated or dislocatable. Newborns have **ligamentous laxity** (this is due to immaturity of the muscles and ligaments, as well as continued effects of maternal hormones, such as elastin). During the newborn period (under 1 month of age) there can be a spontaneous dislocation and reduction of femoral head. If this persists, pathologic changes can occur such as flattening of the acetabulum, joint capsule tightening, and muscle contractures.

Severity of Developmental Dysplasia of the Hip (DDH)



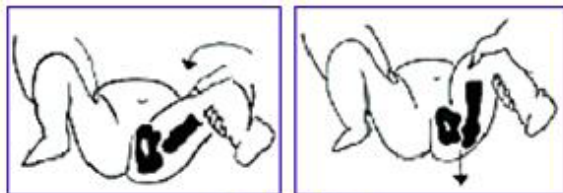
Types of hip dysplasia

Hip dysplasia has a wide range of severity. In some children the ligaments around the hip joint are loose allowing the hip to subluxate. This is when the ball is no longer centered in the socket. Other times the ball is slightly or completely dislocated from the socket.

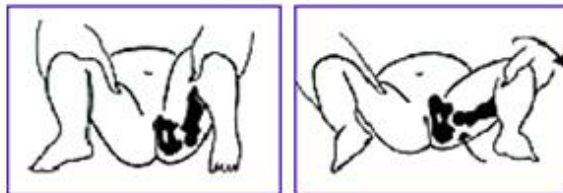
Physical Examination for DDH

Every newborn and toddler has their hips examined at every well child visit. Patients who have developmental hip dysplasia will present with the following:

- Asymmetrical thigh and gluteal folds



Barlow Test



Ortolani Test

● **Barlow:** Place the infant supine, flex the hip & knee, adduct thigh with gentle downward pressure (dislocate). Perform the test when child is less than 2-4 months old.

● **Ortolani:** Place the infant supine, flex the hip and knee, abduct thigh with gentle upward pressure (reduce). Perform the test when child is less than 2-4 months old.

- **Galeazzi:** Place the infant supine, flex the hip and knee and have feet flat on table, examine knee heights for symmetry. Perform the test when child is older than 3 months old.



Galeazzi Test
Difference in knee height

This YouTube video demonstrates the Barlow & Ortolani exams:

<https://www.youtube.com/v/imhl6PLtGLc>

Risk factors for DDH

The following are important risk factors for DDH:

1. Female (9:1)
2. Family history (20%)
3. Breech presentation*
4. Firstborn child (60%)*
5. Oligohydramnios*

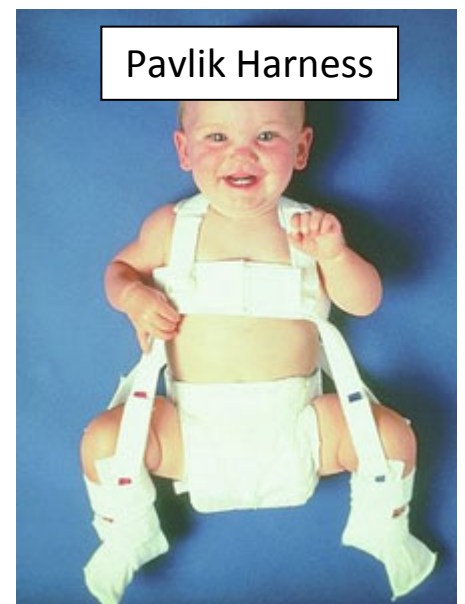
*Anything that restricts the movement of the fetus can lead to abnormal development of a fetus' hip joint.

Diagnosis of DDH

Diagnosis is based on the **physical exam maneuvers** described above. If any of those are positive, or if an infant has risk factors, then imaging studies are ordered in order to confirm the diagnosis. Since the bones of an infant **less than 4 months** of age are not yet ossified, a **hip ultrasound** is the most important imaging study. After ossification has occurred (**older than 4 months**), then a **bilateral hip x-ray** is the best imaging study.

Treatment of DDH

A **Pavlik Harness** is used in children less than 6 months of age. A Pavlik harness keeps the femoral head in the hip socket, promoting proper development of the joint. If the child is older than



6 months, treatment includes a **closed reduction and hip spica cast**. (the spica casts keeps the hips in a similar position to the Pavlik harness, although on a more permanent basis). If closed reduction fails, an open reduction is the next option.

Complications of DDH include avascular necrosis of femoral head and pressure ulcers

Transient Synovitis (aka Toxic Synovitis)

Transient Synovitis is a **diagnosis of exclusion** (in particular it is important to rule out dangerous infection diagnoses like septic arthritis and osteomyelitis). The general

Important tip:

Hip pain often refers to knee

(a patient with hip pathology may complain of knee pain)

presentation is a 3-8 year-old (mean is 6 years old) male (2x more likely to occur in males compared to females) with an **acute onset** of joint pain (hip/knee) and a **painful limp**. These patients are generally afebrile and this presentation is **preceding a URI** (70%).

Diagnosis of Transient Synovitis

Physical examination, labs and x-rays are normal.

BACTERIAL INFECTION *	TRANSIENT SYNOVITIS
Fever—temperature >38.5° C	Afebrile
Leukocytosis	Normal WBC count
ESR >20 mm/hour	Normal ESR and CRP
Refusal to walk	Painful limp
Hip held in external rotation, abduction, and flexion	Hip held normally

* Septic arthritis, osteomyelitis

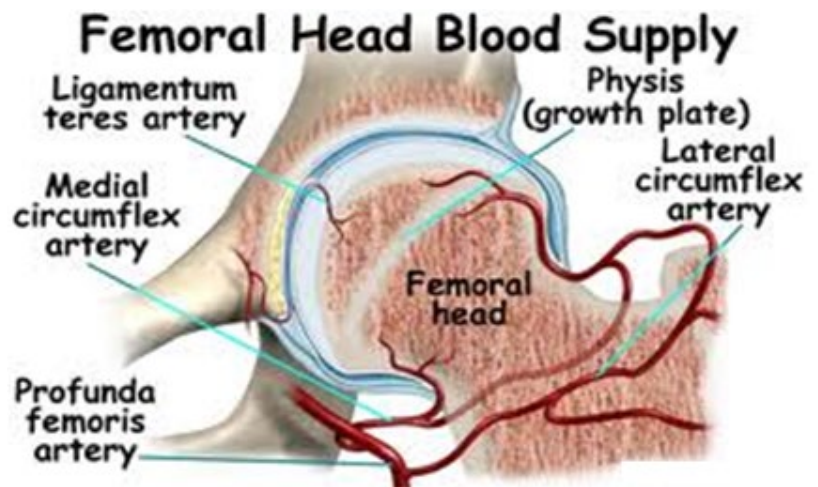
Treatment of Transient Synovitis

Treatment of Transient Synovitis includes **bedrest with minimal weight bearing** and NSAIDS.

Legg-Calve-Perthes Disease

Legg Calve-Perthes Disease is an **idiopathic avascular necrosis of the femoral head**. It's etiology is unknown and could possibly be due to an interruption of blood supply to capital femoral epiphysis. The general

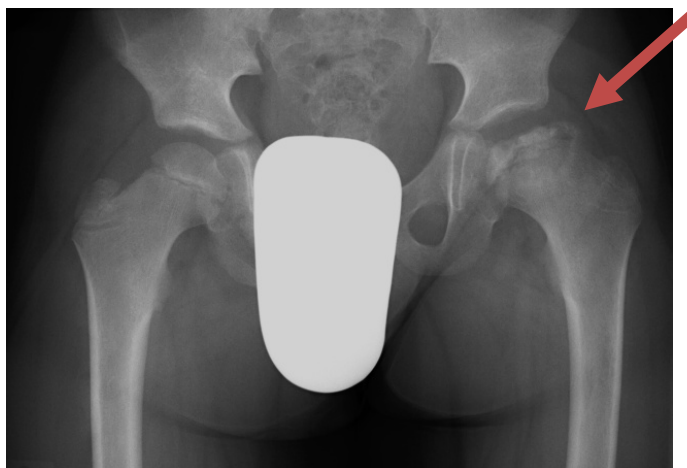
presentation is a **3-12-year-old** (mean is 7 years old) male (4-5x more common in males than females) with atraumatic, **painless** limp. He or she may have decreased range of motion at hip joint and a thigh muscle spasm.



The diagnosis of Legg-Calve-Perthes Disease is based on **AP and frog-leg x-rays of the hips** (*see images below*).

Treatment of Legg-Calve-Perthes Disease

It is most important to urgently refer this patient to a **pediatric orthopedist**. The orthopedist will review ways to control the pain, restore hip motion, and prevent complications (such as femoral head deformity and arthritis). Containment methods such as casts, orthoses (*see images below*), and surgery can also be utilized.



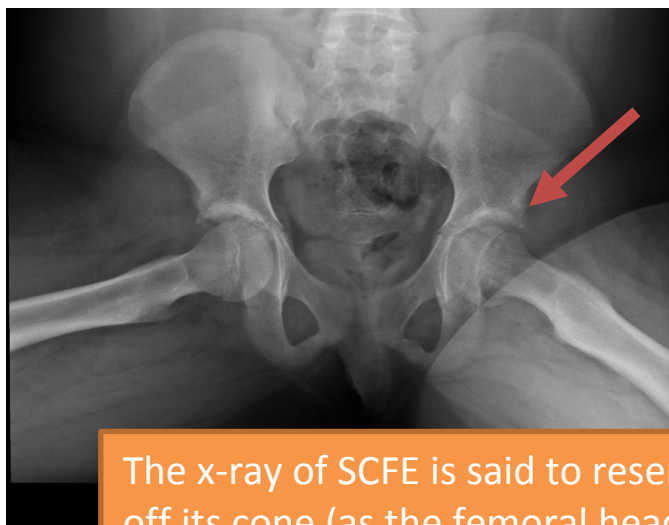
Slipped Capital Femoral Epiphysis (SCFE)

SCFE (pronounced “skiffy”) is the term used for a **femoral head that is displaced, posteriorly and inferiorly on the femoral neck**. This requires **immediate** diagnosis and treatment. The classic presentation is an **obese African American male teenager with acute onset of limp** and hip and/or knee pain. Remember that hip pain often refers to the knee.

During the **physical exam**, the patient may be able to hold their **hip in passive external rotation** and the often refuse to bear weight.

To make a diagnosis, send the patient **via a wheelchair emergently** for **AP and frog-leg** x-rays of the hips. A physician could be held liable if they suspect SCFE, but do not make effort to protect the child’s hip by placing them in a wheelchair until definitive diagnosis has been made,

The AP radiograph: Normally there will be a Klein line drawn straight up the superior aspect of the femoral neck that intersects the epiphysis & femoral head. If such a line does not intersect the epiphysis/femoral head, then it is likely a SCFE.



Treatment of SCFE

A patient will need an **emergent surgery** (internal fixation). Complications include increased severity of SCFE, early degenerative arthritis, and avascular necrosis.

The x-ray of SCFE is said to resemble an ice cream slipping off its cone (as the femoral head slips off the femur)

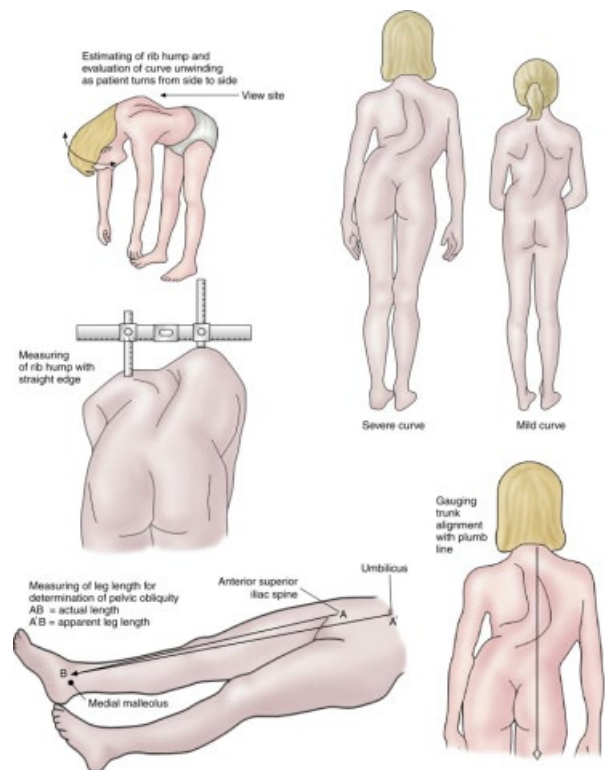
The Spine

Scoliosis

Scoliosis is a condition where there are alterations in normal spinal alignment that occur in the coronal plane. There are 4 types of scoliosis. The first and **most common** is **idiopathic**. Congenital scoliosis presents abnormalities of the vertebral formation in the 1st trimester. Neuromuscular scoliosis is progressive and occurs with cerebral palsy, muscular dystrophy, and spina bifida. Lastly, there is compensatory scoliosis occurs in adolescents with leg-length discrepancies.

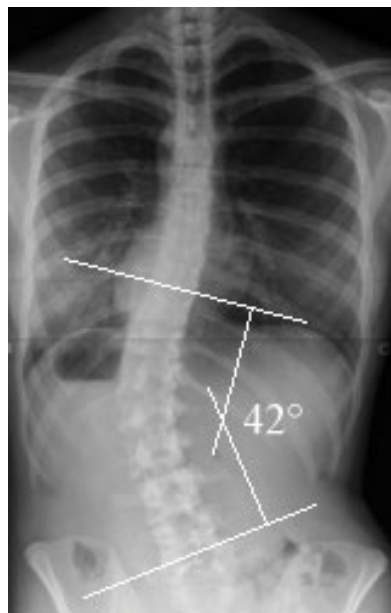
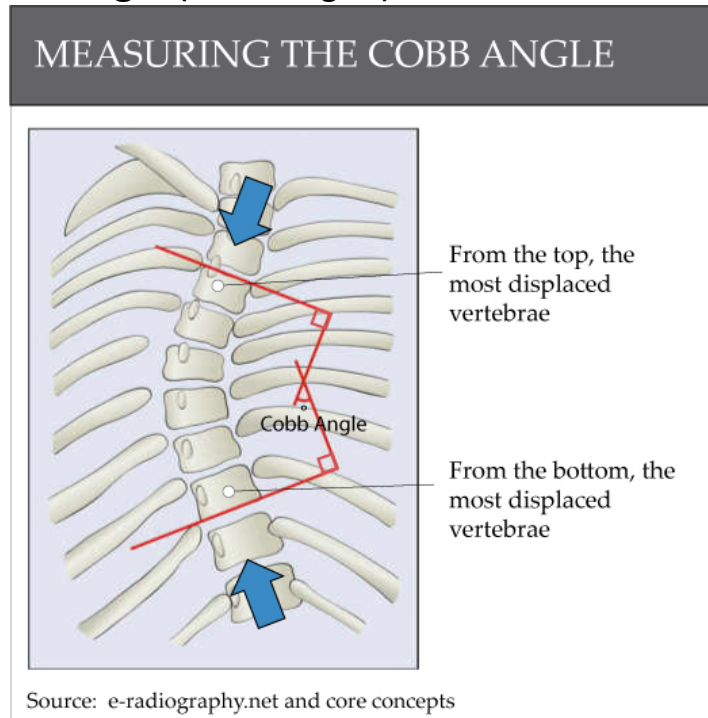
The **diagnosis** is made by having the patient stand with their feet together and **examining their back from behind**. Check for **pelvic asymmetry** and **cutaneous lesions**. Palpate for **tenderness** and then perform the **Adam's forward bend** test (have the patient bend forward to touch their toes, and monitor for asymmetrical rise of the rib cage).

This YouTube video demonstrates the Adam's Forward Bend Test:



https://www.youtube.com/v/n_zHzbdsEI

If any of the above tests are positive, order a **scoliosis series** of x-rays: PA and lateral x-rays of the entire spine. After the films are produced, measure the **Cobb angle** (*see images*).



Before Treatment

July 2009
16-years-old
42°



After 4-weeks of ScolioGold

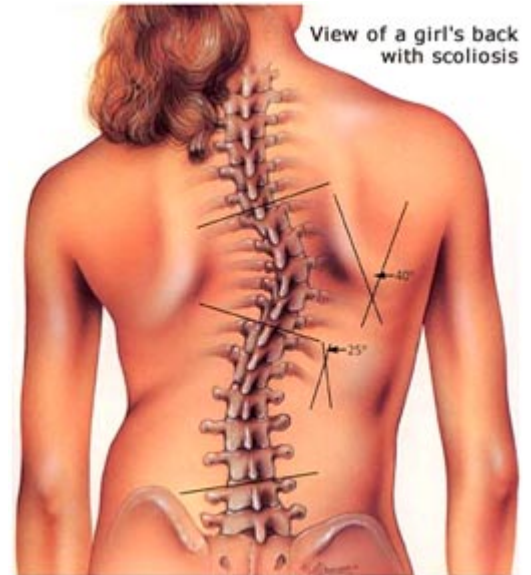
January 2010
17-years-old
28°

Idiopathic Scoliosis

Idiopathic scoliosis occurs in healthy, neurologically normal children. About 20% of cases will have a family history of idiopathic scoliosis. This condition occurs more in females compared to males and is more progressive in females.

There are three main forms of idiopathic scoliosis:

- Infantile (birth-3 years)
- Juvenile (4-10 years)
- Adolescent (>11 years)



Right thoracic curvature is the most common presentation (occurring about 80% of the time) and most patients(70%) do not feel pain.

If the patient has a left-sided curve, an MRI should be order to evaluate for possible intraspinal pathology (ie, a possible tumor or syrinx).

Treatment of Idiopathic Scoliosis

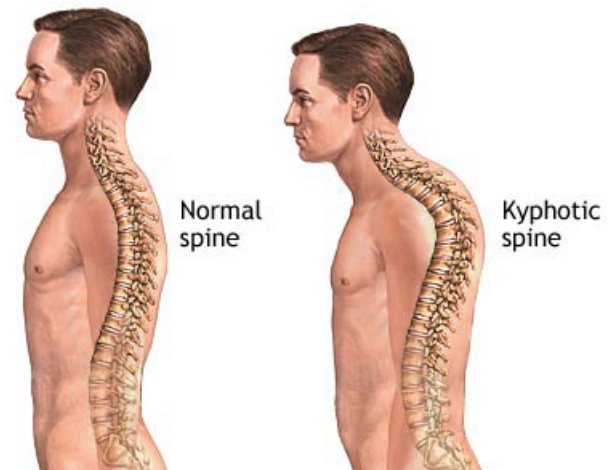
To treat Idiopathic scoliosis several methods can be used depending on the severity and maturity of the patient. **OMT** can be used in all scenarios. Typically if the patient has a Cob angle less than 25 degrees, they are to be observed. If the angle is between 20-50 degrees and the patient is skeletally immature, **bracing** is an option. If the angle is greater than 50 degrees (or with any **pain**), **surgery** can be considered.



Kyphosis

Kyphosis is the **increased angulation** of the thoracic or thoracolumbar spine in the sagittal plane. There are 3 types of kyphosis:

- (1) Postural kyphosis is secondary to poor posture and most often voluntary corrects itself
- (2) Structural kyphosis is also known as Scheuermann's Kyphosis
- (3) Congenital kyphosis is the failure of formation of all or part of the vertebral bodies



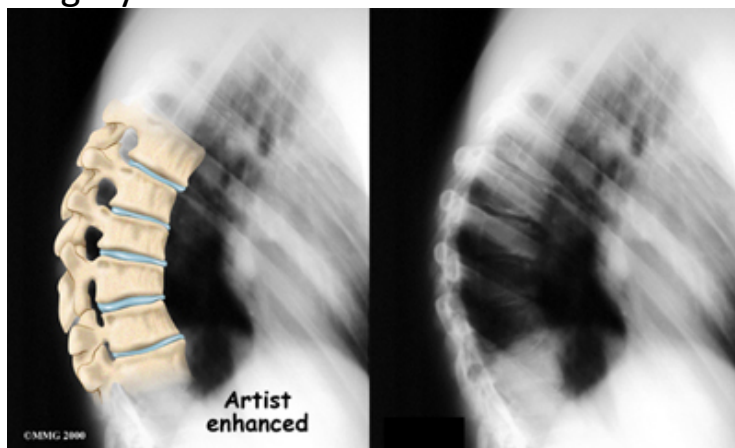
Scheuermann's Kyphosis

Scheuermann's Kyphosis is equally common in males and females. About 50% of cases have back pain. This type of kyphosis **does not correct with standing or lying down**.

To **diagnose Scheuermann's kyphosis**, have the patient bend forward. There will be an **abrupt angulation** and smooth contour in the postural roundback.

X-rays will show a **narrow disk space and anterior wedging**.

Treatment of Scheuermann's kyphosis includes OMT, bracing, and surgery.



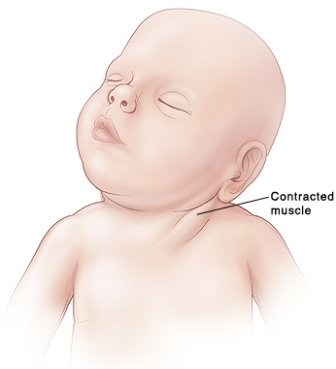
Torticollis

The general presentation of torticollis is a **newborn with a head tilt**. There are two types:

(1) **Muscular torticollis**, in which **both sternocleidomastoid muscles are shortened**.

This could be a result of birth trauma or in utero positioning. Many of these patients have a **palpable mass** or swelling in the sternocleidomastoid muscle.

(2) **Acquired Torticollis** could be due an **abnormality in the upper cervical spine** or a **CNS pathology** (mass lesion)



The **diagnosis** for this condition is **clinical**. AP and lateral cervical spinal films can be ordered to rule out an underlying bony abnormality (but are rarely indicated). An MRI of the head or neck is indicated for persistent pain, focal neurologic symptoms, or persistent deformity.

The **treatment of torticollis** includes **OMT** (muscle energy and soft tissue are the mainstays), **stretching**, and **surgery with post-op OMT/physical therapy**.



Exercise 1. Child undergoing exercise with face looking toward shoulder.



Exercise 2. Child undergoing exercise with ear tilting toward shoulder.

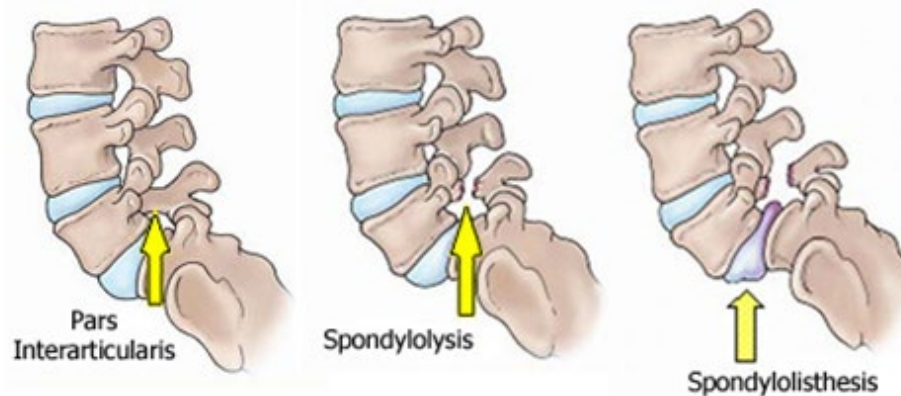
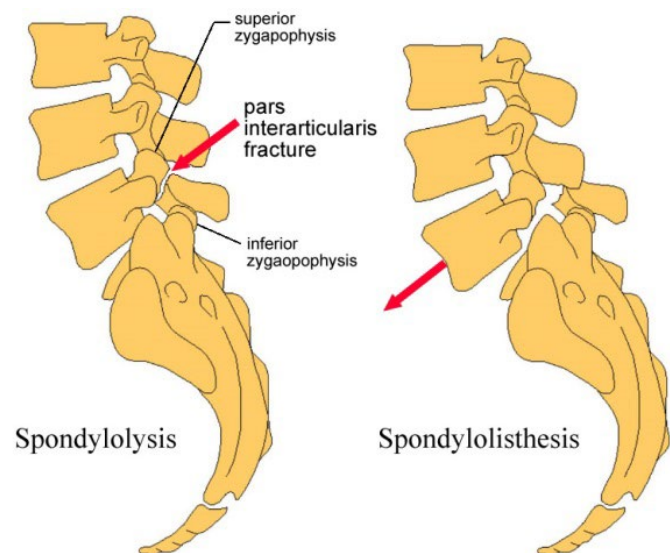
Spondylosis and Spondylolysis

Spondylosis

Spondylosis is a defect in the **pars interarticularis**. This is most common at **L5 or L4**.

Spondylolysis

Spondylolysis is the **bilateral pars interarticularis defects with anterior slippage of the superior vertebra on the inferior vertebra**. This is most common at L5 on S1.



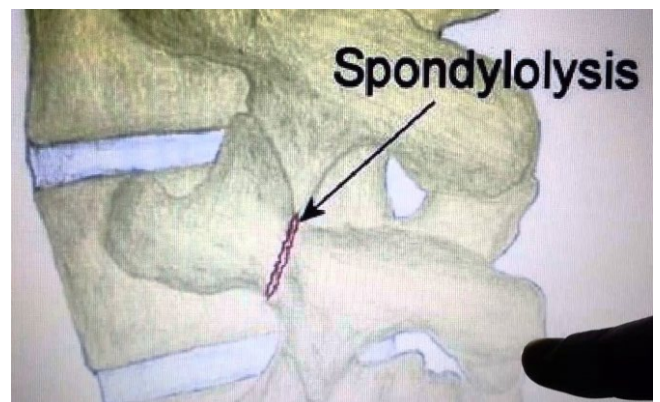
For both Spondylosis and Spondylolysis, the general presentation is an **adolescent athlete** (who plays a sport with **repetitive back extension** such as gymnastics, diving, football, soccer, basketball) with insidious onset of **low back pain for 2 weeks**. The **pain worsens with activity**, may radiate to buttocks, and improves with rest.

Physical Examination of Spondylolysis/Spondylolisthesis

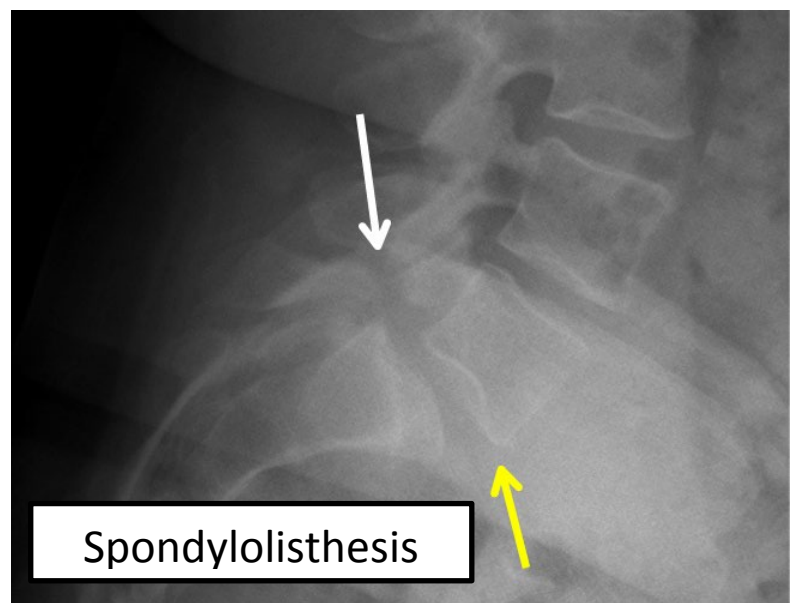
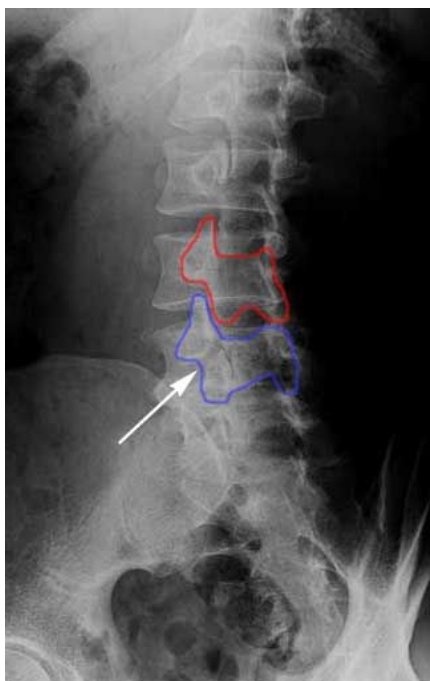
There will be **tenderness to palpation** of the affected vertebrae and, in the case of spondylolisthesis, a **palpable step-off in the LS area**.

A detailed neuro exam must be done to rule out any possible neurologic compromise.

Confirm the **diagnosis** with AP, lateral, and oblique spine x-rays; and CT scans of the affected area.



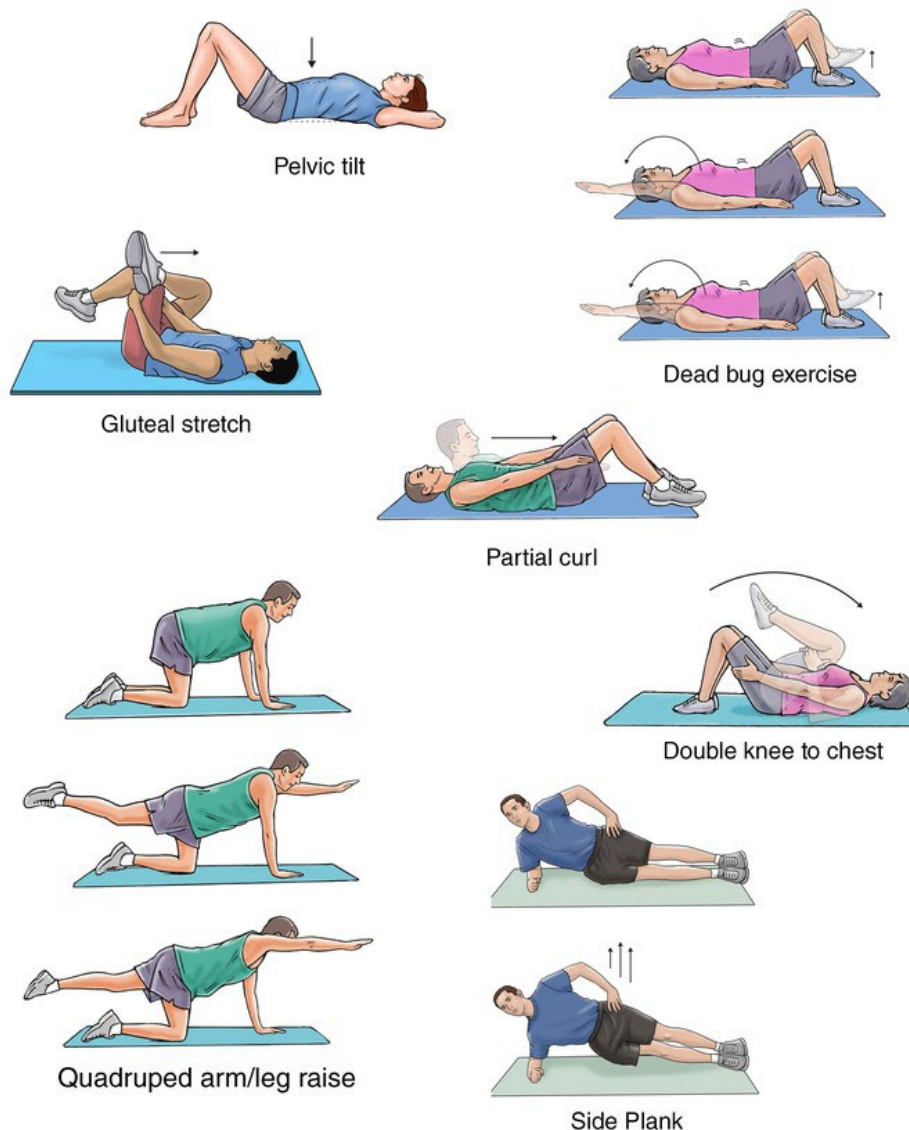
On imaging, spondylolysis is known as the "collar on the Scotty Dog"



Treatment of Spondylolysis/Spondylolisthesis

Treatment depends on the severity. **Activity restriction** is advised for all patients. **OMT/PT** improves lower extremity flexibility, increases core strength, and improves spinal stability. The patient should begin with flexion exercises. **A spinal fusion surgery** is an option if the pain is severe, ROM is greatly restricted, or slippage is progressing.

Spondylolysis and Spondylolisthesis Rehabilitation Exercises



Diskitis

Diskitis is an **intervertebral disk space infection** that does not cause associated vertebral osteomyelitis. It is usually caused by ***staphylococcus aureus***.

The general presentation is a **child younger than 6 years old with back pain**, abdominal pain, irritability, refusal to walk/sit. The patient may or may not have a fever.

The increased vascularity of the intravertebral disks of children makes them more susceptible to this infection than adults are.

A **diagnosis** is made based on the patient's **refusal to flex their lumbar spine** (unusually accompanied by a paravertebral muscular spasm). AP, lateral, and oblique spine x-rays will show a narrow disk space. An **MRI** will help discern between diskitis and vertebral osteomyelitis.



Treatment for Diskitis

Treatment includes **IV antibiotics for 1-2 weeks followed by an additional 4 weeks of oral antibiotics**.

The Upper Extremity

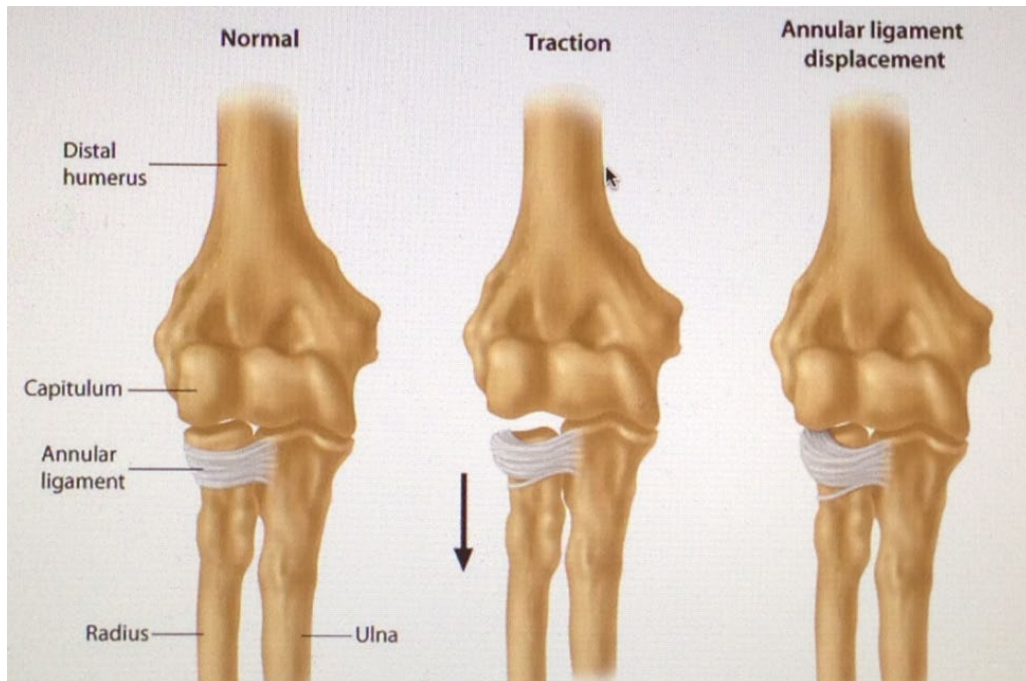
There are a number of orthopedic considerations involving the upper extremity in the pediatric population (*see table below*). For the purposes of this presentation we will be focused on one of the most common, Radial Head Subluxation.

Name	HPI	Dx/Tx
<u>Sprengel</u> Deformity	Congenital elevation of the scapula ; usually unilateral	Mild = reassurance Moderate-severe = surgery
Brachial Plexus Injuries	Either newborn (obstetric brachial plexus palsy), or Athletic injury (lateral flexion of neck away from UE; direct impact to BP; neck extension & rotation toward UE)	Athletic: Usually unilateral and resolve within 15 minutes Must assess cervical spine for serious injury
Proximal humeral epiphysiolysis (<u>aka Little Leaguer's shoulder</u>)	9-14 yo baseball pitcher with pain during/after throwing	X-rays, rest, OMT & strengthening exercises

Radial Head Subluxation (Nursemaid's Elbow)

Radial Head Subluxation is caused most commonly by a **quick pull on the extended elbow of a small child**. The general presentation is an **infant or toddler who holds their hand and distal upper extremity in a pronated position** and refuses to move their hand or elbow.

The **diagnosis** is **clinical**. An x-ray can be ordered if there is a suspected fracture.



Treatment of Radial Head Subluxation

One common treatment is to supinate the patient's hand while applying pressure to radial head. There can be a high rate of recurrence.

Be Careful When U Hold or Carry Your Child Hand
www.facebook.com/learningpetals

The collage includes several key elements:

- Top Left:** A cartoon illustration of a child being pulled by the arm.
- Top Middle:** A photograph of a baby crying while being held.
- Top Right:** A diagram showing the dislocation of the radio head (Dislocación de la cabeza del radio) and the relationship between the humerus, radius, and ulna.
- Bottom Left:** Silhouettes showing incorrect lifting techniques (pulling by the arm or wrist) and a correct technique (lifting from the back).
- Bottom Middle:** A photograph of a crying child being held, with a diagram of the elbow joint showing the dislocated radio head.
- Bottom Right:** A diagram of the elbow joint with labels for the humerus, radius, ulna, and annular ligament. Text states: "Solo el médico debe acomodar el ligamento que sale de su lugar" (Only the doctor should reposition the ligament that has come out of its place).

Don't Forget To Share

Additional text in the collage includes: "No levantar al niño tomándolo de las mano o de las muñecas" (Do not lift the child by the hands or wrists) and "Luego de la intervención médica, la recuperación es inmediata" (After medical intervention, recovery is immediate).

This YouTube video demonstrates reduction of radial head subluxation:
<https://www.youtube.com/v/tJb5rGOFiTY>

References

Kliegman RM. *Nelson: Essentials of Pediatrics*. 7th Edition. Saunders; 2014

- Chapters 199-203

- Text available online via TUCOM library (Call# RJ45)

<https://www.youtube.com/v/imhl6PLtGLc> = Barlow & Ortolani Tests

https://www.youtube.com/v/n_zHzbdsEI = Adam's Forward Bend Test

<https://www.youtube.com/v/tJb5rGOFiTY> = Reducing a Nursemaid's Elbow

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<https://quizlet.com/61140828/pediatric-musculoskeletal-disorders-flash-cards/> =
Pediatric Musculoskeletal Disorders Part 2